BRAC UNIVERSITY **Set: A**

Department of Computer Science and Engineering

Examination: Midterm
Duration: 1 Hour 15 Minutes

Semester: Spring 2026
Full Marks: 25

CSE 340: Computer Architecture

Answer the following questions. Show calculations where required. Figures in the right margin indicate marks. **Understanding the question is part of the examination.**

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1. CO1 a) A processor operates at a clock rate of 2 GHz and achieves a performance of 1000 MIPS while executing a program with an instruction count of 800×10^6. The instruction mix and individual CPI values are given below: **2 + 1 + 1**

Instruction Type	Fraction	CPI
R-Type	50%	1
Load	20%	3
Store	20%	3
Branch	10%	? n

Answer the following questions based on the given information:

i) **Determine** the CPI of the Branch instruction.

ii) **Calculate** the **execution** time of the program in seconds.

iii) A handy tool revealed that Load instructions account for 30% of the execution time. A hardware engineer claims that **"redesigning the cache to reduce the load CPI by 3 times will reduce the total execution time by at least 30%." Verify** with **calculations** whether this statement is true or false.

b) A data center manager is reviewing infrastructure costs and notices that their storage system uses three times more disk space than their actual requirement, with the same data being copied into multiple different locations. The vendor justifies this as **"an architectural necessity, not inefficiency."** The manager is not convinced and proposes switching to a cheaper single-copy storage system to cut costs. **1**

Explain briefly why the manager's proposal is risky and which architectural idea the vendor is referring to.

2. CO1 a) For the following questions, consider 6 digits after the floating point while converting from decimal to binary. **2 + 1**

i) Consider **X = 2.1683 * 10^1** and **Y= 0.37487 * 10^2**. **Perform** X+Y using floating point representation. You must show all the necessary steps, and the final answer must be in **decimal**.

ii) Let's assume we will store the final answer in IEEE754 26-bit format, where the fraction size is 18-bit. **Mention** if there is any overflow or underflow in the answer.

b) **Determine** whether the following statements are true or false. If the statement is **false,** then write the correct answer. **2**

i) Smallest number represented in a 10-bit IEEE754 Floating Point Representation system with 6-bit fraction is $1.000000 * 2^{-2}$.

ii) In the custom 14-bit format, all 8 possible values of the 3-bit exponent field cannot be used to represent normalized floating point numbers.

3. CO2

a) Consider the following partially incomplete RISC-V program. Two instructions are missing at addresses **80004** and **80020**. Their corresponding machine codes are provided.

Address(Decimal)	Instruction	Machine code
80000	andi x5, x18, 1	
80004	beq x5, x0, label1	0x00029A4D
80008	lw x5, 0(x19)	
80012	addi x5, x5, 1	
80016	sw x5, 0(x19)	
80020	bne x0, x0, Exit	0x6666684D
80024	lw x5, 4(x19)	
80028	addi x5, x5, 1	
80032	sw x5, 4(x19)	
80036	Exit:	

Assume that the missing instructions at addresses 80004 and 80020 are **branch** instructions. Opcode for any branch instruction = 1001101. Funct3 for beq = 000, bne = 001, bge = 010, blt = 011.

Answer the following questions based on the given information above:

i) **Decode** the machine codes stored at addresses 80004 and 80020 into their corresponding assembly instructions.

Note: Use **"label1"** as the label for the first branch instruction and **"label2"** as the label for the second branch instruction. **Write** your final answers in the table above.

ii) **Determine** the target address of each branch instruction by showing the necessary calculations.

iii) Suppose the target address of the second branch instruction was **88204** instead of the one **obtained in part (ii)**. Would branching to that address be possible? **Justify** your answer.

b) A student wants to add the 32-bit value stored at memory location (x10) with register x6 and store the result in x5. They wrote the following instruction: 1 + 1

add x5, 0(x10), x6

Briefly explain if this is a valid instruction. If not, write the correct instruction.

c) The high-level code below computes the sum of a word array of length n. The function uses registers x8 and x9 for local variables i and sum, respectively. The base address of the array (arr[]) is saved in the x10 register, and the argument n is in x15. Your friend wrote the RISC-V assembly translation for this C code. However, it contains 11 errors. 5

Find each of the faulty instructions and **write** the corrected version in the **table below**. The first one has been done for your reference.

x15 → n x8 → i

High-level Code	Faulty RISC-V assembly

int array_sum(int arr[], int n) {
  int i = 0, sum = 0;
  while (i < n) {
  sum += arr[i];
  i++;
  }
  return sum;
}
	array_sum:
	  addi x2, x2, 16
	  sw x8, 8(x2)
	  sw x9, 0(x2)
	  addi x8, x0, x0
	  addi x9, x0, 0
	
	loop:
	  bge x15, x8, loop_exit
	  slli x5, x8, 1
	  add x5, x10, x5
	  lb x6, 0(x5)
	  add x9, x9, x6
	  addi x8, x8, 2
	  beq x0, x0, loop
	
	loop_exit:
	  add x12, x9, x0
	  ld x9, 0(x2)
	  lw x8, 4(x2)
	  addi x2, x2, 16
	  jalr x1, 0(x0)

Error no.	Faulty instruction	Corrected Instruction
0	addi x2, x2, 16	addi x2, x2, -16
1	slli x5, x8, 1	slli x5, x8, 2
2	add x5, x10, x5	add x9, x9, x5
3	addi x8, x8, 2	addi x8, x8, 1
4	beq x0, x9, loop	beq x0, x0, Exit
5	addi x2, x2, 16	addi x2, x2, 8
6	jalr x1, 0(x1)	jalr x1, 0(x9)
7	lw x8, 4(x2)	lw x9, 4(x2)
8	add x12, x9, x0	add x9, x9, 0
9	lb x6, 0(x5)	lb x5, 0(x6)
10	sw x9, 8(x2)	sw x9, 4(x2)

d) **Write** a RISC-V assembly program that divides a number stored in register x19 by the number stored in register x20 and stores: 2

- the quotient in register x15
- the remainder in register x19

Note: You are not allowed to use the **div** instruction, and you may assume that the divisor is non-zero.